1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L37 – Agile Development**

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3 **SDLCs**

- We have seen two (water-fall, ISLC)
- There are others (spiral, iterative, evolutionary …)
- These are all "traditional" models
  - Stages are mostly the same with variations in execution sequence
  - Equally rigorous with strict requirements at each stage
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5 **Recap**

- Requirement stage – functional and usability (CI)
- Design– design guidelines (usability), DFD+ER, OOD+UML
- Prototype (including evaluation) – CW, HE
- Coding and code testing – black-box, white-box, integration + system testing
- Usability testing

6 **Recap**

- Requirement stage – functional and usability (CI)
  - How long it takes?

7 **Recap**

- Interface design
  - Design – prototype-evaluate cycle
  - Design guidelines + brainstorming
  - Prototype building
  - Evaluation – CW, HE
  - How long it takes?

8 **Recap**

- System design
  - DFD+ER/OOD+UML
  - How long it takes?

9 **Recap**

- Implementation (coding)
  - Unit level coding + testing – review-based, black-box, white-box
  - Integration + system testing

- How long it takes?

10 **Recap**

- Usability testing
  - How long it takes?

11 **Problem with Traditional SDLCs**

- Rigidity – may lead to time and cost overrun (and even non-acceptance by customer)
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12 **Why Problematic**

- Due to strict requirements at each stage
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13 **Why Problematic**

- All requirements to be identified first
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- Necessary to implements ALL requirements BEFORE deployments …
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14 **Why Problematic**

- Every stage detailed documentation (lots of documentations!)
  - What are the documents to be generated in our ISLC?
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15 **Documents**

- Let's list those

16 **Agile Development**

17 **Idea**

- Agile (fast) development – ensure quick turn-around of projects
  - Eliminate unnecessary things and documentations
  - Start small (don't try to do everything from start)
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18 **Agile Process Flow (A-SDLC)**

19 **Agile Iteration Workflow**

- Typically of 4-6 weeks
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20 **How to Make it Work**

- Daily meetings (of the development team)
- Live demo – after each iteration
- Feedback – regularly collect from stakeholders/customers and share with team before next sprint
- Remain agile – be open to make changes based on feedback

21 **Some Agile Methods**

- Scrum
- Extreme programming (XP)
- Disciplined agile delivery (DAD)
- Dynamic systems development method (DSDM)
- …

22 **Further Reading**

- Agile manifesto (https://agilemanifesto.org/)
- 12 agile development principles (https://www.agilealliance.org/agile101/12-principles-behind-the-agile-manifesto/)